Physics Letters A ••• (••••) •••–•••

Contents lists available at ScienceDirect

Physics Letters A

www.elsevier.com/locate/pla

Areas and sizes of cascades in dissipative one-dimensional sandpile model

Jie Liu, P.M. Hui *

Department of Physics, The Chinese University of Hong Kong, Shatin, Hong Kong

ARTICLE INFO

Article history:
Received 2 February 2017
Received in revised form 11 May 2017
Accepted 12 May 2017
Available online xxxx
Communicated by C.R. Doering

Keywords:
Dissipative sandpile model
One-dimensional systems
Distribution of cascade areas
Distribution of cascade sizes
Self-organized processes

ABSTRACT

A prototypical self-organized sandpile mode is studied on a one-dimensional (1D) chain with periodic boundary conditions. A dissipation mechanism in which every grain being transferred between nodes has a probability ϵ to be taken out is needed, as the system has no boundary nodes for grains to fall off – a feature reminiscent of complex networks. Detailed numerical analysis revealed distributions of cascade areas $D(a)$ and cascade sizes $D(s)$ that are intrinsically different from other 1D sandpile models with an open end. Analyzing cascading processes on a chain, independent-site approximations to $D(a)$ and $D(s)$ are developed. The approximated distributions are given in terms of a single parameter ϕ_0, which is the fraction of empty nodes when the system is stable. The approximations are shown to capture the key features of the distributions. The distribution of cascade sizes $D(s)$ is shown to exhibit large fluctuations that cannot be suppressed by averaging over different realizations. Our approximation provides a physically transparent explanation of the intrinsic large fluctuations in terms of the number of ways that a cascade can proceed for achieving a certain size. To close the approximations, a semi-empirical formula for the parameter ϕ_0 as a function of the dissipation probability ϵ is found. Our work thus reports non-trivial results on a seemingly simple model and sheds light on analyzing cascading processes in other complex networks with no boundary nodes.

© 2017 Elsevier B.V. All rights reserved.

UNCORRECTED PROOF

1. Introduction

Since the proposal of self-organized critical phenomena within the context of a sandpile model in the late 1980's, the topic has continued to attract much attention and grow into an important research area due to its wide range of applications. The development of self-organized criticality (SOC) into a research field was summarized by the original proposer [1]. Exhibiting spatial and temporal scale invariance, SOC is conceived to be the underlying mechanism for a wide range of natural and social phenomena, such as earthquakes [2], solar flares [3], blackouts of electric power systems [4] and financial crises [5]. Its importance grew stronger when more empirical results [6–8] were found to support its relevance. Excellent surveys on the key concepts of the field were given by Jensen [9] and Christensen and Moloney [10].

A prototype model for studying SOC is the sandpile model of Bak et al. [11,12] (henceforth referred to as the BTW sandpile model). Many interesting variations of the model were proposed and studied [13–20]. The model was initially studied on regular lattices. Resembling the dynamics of a real sandpile, grains can fall out of a lattice from the sites at the boundary. This provides a natural mechanism for dissipating the grains and achieve stability. The rapid developments in complex networks also prompted many studies of SOC on such networks. These networks have different topological characteristics from regular lattices [21–24], especially in the inhomogeneous connectivity among the nodes. Understandably, the inhomogeneity would affect how the grains toppled from one node affect another. As a result, the BTW sandpile model has been studied in different networks such as random graphs [25–28], scale-free networks [29,30] and inter-dependent networks [31]. In contrast to regular lattices, these networks have no boundary nodes for the grains to fall out of the system. A dissipation mechanism is therefore invoked [29,26,27] to take grains away from the system. Typically, dissipation is introduced through a probability of removing a grain every time when a grain is transferred from one node to another due to toppling inside the system. This dissipation mechanism, which has been used extensively in numerical studies of SOC in complex networks, but a good understanding of the effects of the dissipating mechanism is lacking. The problem turns out to be non-trivial. While critical behavior of sandpile models with the dissipation mechanisms were observed and studied on regular lattice, small-world networks and random networks [32–35], the behavior in a one-dimensional chain could be very

* Corresponding author.
E-mail address: pmhui@phy.cuhk.edu.hk (P.M. Hui).

http://dx.doi.org/10.1016/j.physleta.2017.05.022
0375-9601/© 2017 Elsevier B.V. All rights reserved.

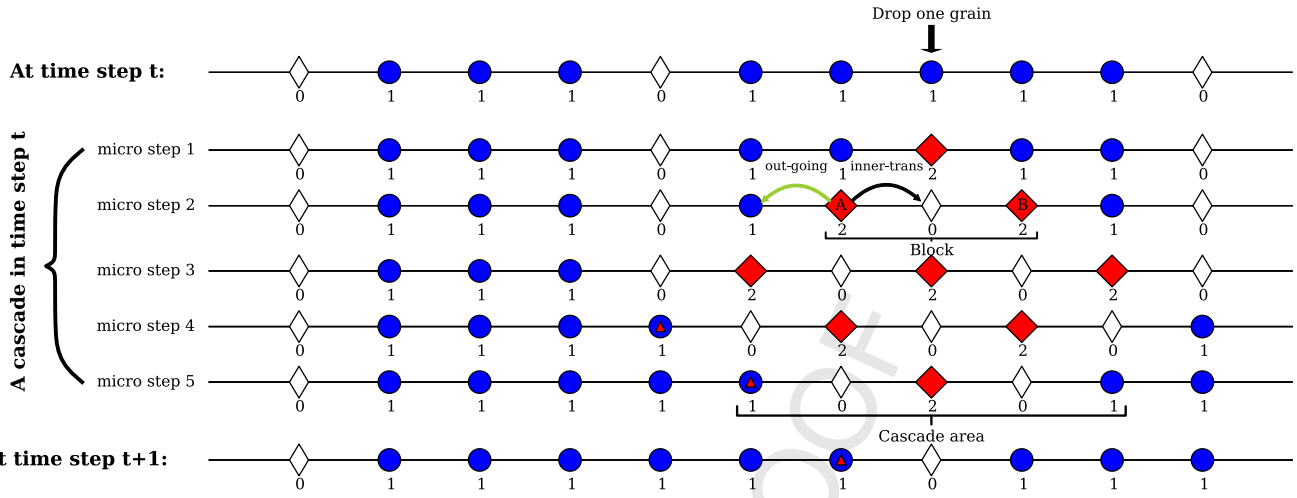


Fig. 1. Schematic diagram showing the dynamics in a segment of a one-dimensional chain that goes from one stable configuration at time step t to another stable configuration at time step $t+1$ through a number of toppling processes that occur in the micro time steps. A stable configuration (top row) consists only of load-0 nodes (open diamonds) and load-1 nodes (closed blue circles). When a grain is dropped on a load-1 node, it becomes an unstable load-2 node (red closed diamond) in micro time step 1. This triggers a toppling that turns the two neighboring nodes into load-2 nodes (see A and B nodes in micro time step 2). These nodes topple and lead to three unstable load-2 nodes at micro time step 3. The expansion of the continuous block of cascading nodes ends when the grain at the end of the block falls onto an empty exterior site and turns it into a stable load-1 node (see micro time step 4). A toppling occurs in micro time step 5 within the interior of the block and the system retains stability. The process at time step t ends and time step $t+1$ begins. The expansion of the block takes $a_l = 3$ and $a_r = 3$ micro steps to end, giving a cascade area of $a_l + a_r - 1 = 5$ (as marked) and a cascade size of $a_l \cdot a_r = 9$ (see 9 red diamonds in the micro time steps). Note that had it been $a_l = 4$ and $a_r = 3$ at time step t (open diamond on the left moves to the left by one node), there would be three more cascades at the nodes marked with a triangle inside the closed blue circles (see micro time steps 4, 5 and time step $t+1$), giving a total of 12 cascades as given again by $a_l \cdot a_r$. The cascade area now involves one more node on the left (see micro time step 4) and is given by $a_l + a_r - 1 = 6$ again. (For interpretation of the references to color in this figure, the reader is referred to the web version of this article.)

different and it is usually *not* critical. Here, we take advantage of the simplicity of a one-dimensional (1D) system and study BTW sandpile model in a closed system in the presence of dissipations. It should be noted that some 1D BTW sandpile models exhibit simple behavior [10]. For example, in a 1D system of length L having a wall on one side and an open boundary for grains to fall off on the other end, there is only one stable configuration with a uniform distribution of cascade sizes ranging from size 1 to L. Lahtinen et al. also studied 1D system with open boundary nodes at both ends [36]. However, the basic problem of BTW sandpile model on a 1D chain with periodic boundary, and hence no boundary nodes, and a dissipation mechanism remains to be investigated. Simple as it may seem, we show that the distributions of cascade areas and cascade sizes are intrinsically different from other 1D models with an open end. The distributions are non-uniform, with intrinsic large fluctuations in the distribution of cascade sizes. Different dissipation parameters give rise to different distributions. Studying the dissipation mechanism in 1D systems also has the advantage that the non-trivial features in the distributions can be understood analytically, albeit approximately. Our results, therefore, shed light on how to understand the effects of dissipations in sandpile models on complex networks, which also do not carry boundary nodes.

The plan of the present work is as follows. In Sec. 2, the model is defined and the quantities of interest are introduced. In Sec. 3, simulation results of the distribution of cascade areas $D(a)$ and distribution of cascade sizes $D(s)$ are presented for different dissipation probabilities. We develop approximations for the distributions by ignoring the spatial correlation of the grain-loading on neighboring nodes in the steady states. These approximations are shown to capture the key features of the actual distributions. In addition, we show that the large fluctuations in $D(s)$ are intrinsic and they cannot be removed by averaging over many realizations. Our approximation also illustrates the origin of the large fluctuations. In Sec. 4, we propose a semi-empirical formula for the fraction of empty sites in the steady state, which is the single parameter that enters into the approximated distributions, as a function of the dissipation probability and show that it is in excellent agreement with numerical results.

2. Model

We study a sandpile model on 1D chain of N nodes with periodic boundary condition, with the N-th node connected back to the first node. Each node has a certain capacity, representing the largest number of grains of sand that the node can hold without toppling. The number of grains that a node bears instantaneously is called its load. At the beginning of a time step, a grain of sand is dropped on to a randomly chosen node. When the load at a node exceeds the capacity, the grains topple with each neighboring node receiving one grain. Note that there are no boundary nodes for grains to leak out of the system, a dissipation mechanism is invoked to keep grains from accumulating. For each grain sending out in a toppling, it could be dissipated with a probability ϵ. If it is not dissipated, it will be added to the neighboring node and thus may lead to further toppling. Rather generally, the capacity of a node can be taken to be its degree minus one [29,25,30]. For a chain, the capacity is 1 and it is uniform over all nodes. In fact, the choice of capacity will not alter the results as long as a toppled node can send out a grain to each of its neighboring nodes. It should be noted that all nodes behave in the same way in our model. In contrast, an alternative way to introduce dissipation is to designate a certain selection of sites as sinks at which grains will be taken out of the system.

For later discussion, it is convenient to label a node with i grains a load-i node. On a chain, load-0 and load-1 nodes are stable, but load-2 nodes are unstable and they will topple. A *stable configuration* of the system refers to one in which every node is stable, i.e., only load-0 and load-1 nodes exist. Otherwise, the system is in an unstable configuration in which toppling and/or dissipation occur until a stable configuration is achieved. Fig. 1 illustrates the cascading processes typical of the system in one time step and also the quantities of interest in the present work. At a time step t, one grain is added. If it is dropped on to a load-0 (white) node, the system remains in a stable configuration and nothing happens. In this case, we record a "cascade" of size 0 and of area 0 at time t and move on to the next time step. If the grain is dropped on to a load-1 (blue) node, as shown in Fig. 1, the node

UNCORRECTED PROOF

Please cite this article in press as: J. Liu, P.M. Hui, Areas and sizes of cascades in dissipative one-dimensional sandpile model, Phys. Lett. A (2017), http://dx.doi.org/10.1016/j.physleta.2017.05.022

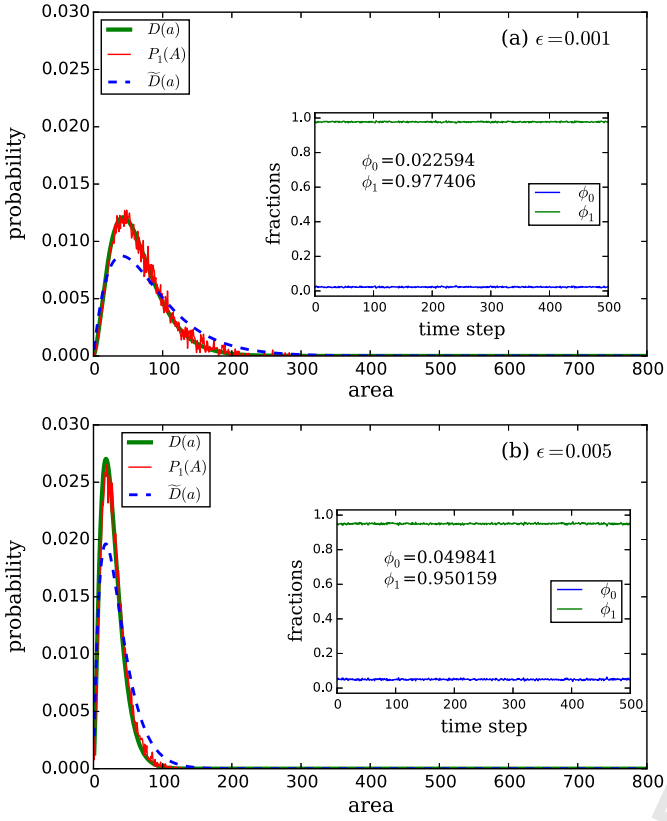


Fig. 2. Distribution of cascade areas $D(a)$ (green solid line) for two different values of dissipation probabilities (a) $\epsilon = 0.001$ and (b) $\epsilon = 0.005$. The approximation $\widetilde{D}(a)$ (Eq. (4)) using ϕ_0 extracted from simulations (insets) is shown for comparison (blue dashed line). The distribution of strings of load-1 nodes in stable configurations $P_1(A)$ extracted from a single realization (thin line) is shown to follow $D(a)$ very well. The values of ϕ_0 and ϕ_1 (insets) are dependent on the dissipation probability ϵ. (For interpretation of the references to color in this figure, the reader is referred to the web version of this article.)

becomes load-2 (red) and the configuration is unstable. The node topples and the grains sending out may trigger further nodes to topple. Fig. 1 schematically shows a case in which the cascade lasts for 5 micro steps during which the system goes from one unstable configuration to another and a string of nodes and a number of toppling are involved before the system settles into a stable configuration. Only then the time step t ends and the next time step begins [15,10,6]. Two quantities are of interest. The *size* s of a cascade at a time step is defined as *number of times* of toppling before the system regains stability after a grain is dropped. The *area* a of a cascade at a time step is defined as the *number of different nodes* that at least a toppling has occurred before the system regains stability. As a node may topple more than once in a time step, $s \geq a$. The features in the distributions of cascade areas $D(a)$ and cascade sizes $D(s)$ and their underlying physics are the focus of this work.

3. Results

Extensive numerical simulations have been carried out on systems with $N = 2000$ nodes. In a realization, the nodes are initially empty and the system is then driven to a steady state by adding grains repeatedly. After a transient period, data are recorded for the cascade size and area in each time step for a long time. To obtain better statistics, the distributions $D(a)$ and $D(s)$ are calculated by averaging data over 500 different realizations. We checked that using systems of bigger sizes and averaging over more realizations do not change the results reported here.

Fig. 2 gives the distribution of cascade areas $D(a)$ (thick green line) for $a \geq 1$ for two dissipation rates $\epsilon = 0.001$ and 0.005. There is no binning of the data. There is a finite $D(0)$ at $a = 0$ for zero cascade area that is not shown in Fig. 2 for clarity. The key features are: (i) $D(a)$ is a smooth function, in sharp contrast with $D(s)$ to be discussed later; (ii) $D(a)$ increases with a for small a and attains a peak before dropping with a longer tail than it rises; and (iii) the value at which $D(a)$ peaks is dependent on the dissipation ϵ.

Note that the choice of system size and values of ϵ is made to avoid finite size effects. From Fig. 2, the probability of having cascades of areas higher than 300 is almost zero for both values of ϵ. This makes sure that the distributions are generic and dependent only on ϵ but not the system size.

To understand how these features arise, let ϕ_0 and ϕ_1 be the fractions of load-0 nodes and load-1 nodes in the stable configurations after the transient, respectively. With probability ϕ_0, a grain will be dropped on to a load-0 node giving a cascade of area $a = 0$. Thus, ϕ_0 also gives $D(0)$. With probability ϕ_1, a grain is dropped on a load-1 node, giving rise to a cascade of area $a \geq 1$. An approximation to $D(a)$ can be developed in terms of ϕ_0.

In a cascade, nodes topple one-by-one along the chain. Therefore, the nodes that undergo toppling through the micro time steps form a continuous block. An example is shown in Fig. 1. After a grain is dropped on a load-1 node at the beginning of time step t, the node becomes an unstable load-2 node (filled red diamond) in micro time step 1. Via toppling, the cascade area may spread and expand to neighboring nodes continuously. In micro time step 2, the two neighboring nodes (labelled A and B) become unstable after receiving a toppled grain. The block of unstable nodes now covers a string of three nodes. The block can continue to expand if the toppled grains from A and B could lead to more unstable nodes at the nodes next to the two ends of the block. Note that there are toppled grains from A and B that fall back to interior nodes inside the block and they will *not* increase the cascade area a. Only the grains that fall on to exterior nodes outside the instantaneous cascading block will make the block longer and increase a (see micro time step 2 to 3). The block ceases to expand in one direction when a grain sending out to an exterior node is either dissipated or falling onto a load-0 node (see micro step 3 to 4).

Assuming that every node is found to be a load-0 node with probability ϕ_0, i.e., ignoring the spatial correlation of load-0 and load-1 nodes, the probability of a grain sending outside a continuous block to be either dissipated or falling on to a load-0 node can be expressed as

$$\epsilon_{eff} = 1 - (1 - \epsilon)(1 - \phi_0) = \epsilon + \phi_0 - \epsilon\phi_0. \tag{1}$$

Let a_l (a_r) be the number of micro time steps for a block to stop expanding to the left (right). Ignoring how load-0 and load-1 nodes are correlated spatially, these quantities follow a geometric distribution with the parameter ϵ_{eff}, i.e., the probability $P(a_l = a')$ for a_l to take on the value a' is

$$P(a_l = a') = P(a_r = a') = (1 - \epsilon_{eff})^{a'-1}\epsilon_{eff}\ . \tag{2}$$

The cascade area a is related to a_l and a_r by $a = a_l + a_r - 1$. In the example illustrated in Fig. 1, the expansion in both directions stops at $a_r = a_l = 3$, giving $a = 5$ (see cascade area marked). Therefore, the distribution of cascade areas can be expressed as

$$D(a) = \phi_0\delta_{a,0} + (1 - \phi_0)\sum_{i=1}^{a} P(a_l = i) \times P(a_r = a + 1 - i). \tag{3}$$

Applying Eq. (2) leads to an approximation to the distribution of cascade areas given by

$$\widetilde{D}(a) = \phi_0\delta_{a,0} + a(1 - \phi_0)(1 - \epsilon_{eff})^{a-1}\epsilon_{eff}^2. \tag{4}$$

Eq. (1) and Eq. (4) rely on one single parameter ϕ_0.

UNCORRECTED PROOF

Please cite this article in press as: J. Liu, P.M. Hui, Areas and sizes of cascades in dissipative one-dimensional sandpile model, Phys. Lett. A (2017), http://dx.doi.org/10.1016/j.physleta.2017.05.022

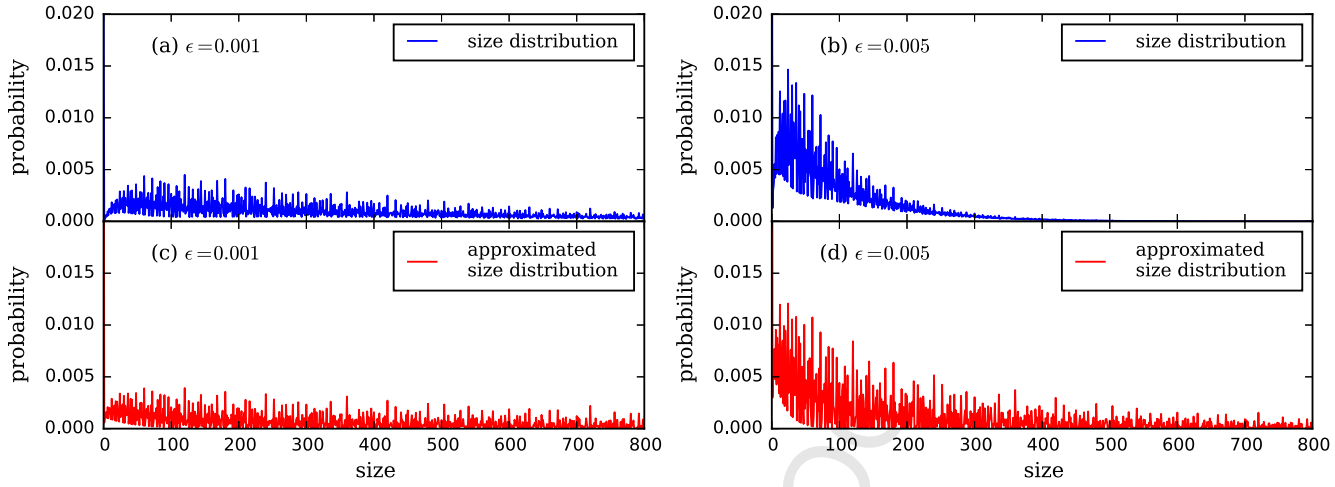


Fig. 3. Distribution of cascade sizes $D(s)$ for two different values of dissipation probabilities (a) $\epsilon = 0.001$ and (b) $\epsilon = 0.005$ as obtained by numerical simulations. Results are averaged over 500 different realizations. The fluctuations in $D(s)$ are intrinsic and they cannot be suppressed by averaging over different realizations. Results of the approximated $\widetilde{D}(s)$ (Eq. (5)) using ϕ_0 extracted numerically are shown in (c) and (d) for comparison.

To check the validity of the approximated distribution and the underlying assumption, we extract the value of ϕ_0 from simulations (see insets in Fig. 2). Substituting the value into Eq. (1) and Eq. (4) gives $\widetilde{D}(a)$, which is shown for comparison in Fig. 2 (blue dashed line). For the two values of ϵ, $\widetilde{D}(a)$ captures the key features of the actual $D(a)$. They agree quantitatively for the size where the distribution peaks. For $\epsilon = 0.001$ and 0.005, the most probable cascade areas are 41 and 18, respectively, for both $D(a)$ and $\widetilde{D}(a)$. There are, however, discrepancies. The approximated distribution $\widetilde{D}(a)$ attains a smaller peak value and drops off with a fatter tail, indicating that the spatial correlation of the states of the nodes is important and the assumption that each node is a load-0 node with probability ϕ_0 is at best an approximation. From another viewpoint, $D(a)$ should be closely related to the distribution of clusters of neighboring load-1 nodes in stable configurations, especially in the limit of $\epsilon \ll 1$ where the cascade area is dominantly governed by the size of cluster of load-1 nodes in which the first cascade is triggered by adding a grain. The example in Fig. 1 shows two clusters of load-1 nodes of sizes 3 and 5 at the beginning of time step t. The load-1 cluster size distribution $P_1(A)$, which is also the probability of dropping a grain on to a load-1 node belonging to a cluster of size a, is therefore a good approximation to $D(a)$ for $\epsilon \ll 1$. The distribution $P_1(A)$ as obtained in a single run is also shown in Fig. 2. It is in good agreement with $D(a)$ for both values of ϵ. Nonetheless, obtaining a closed form for $P_1(A)$ and hence $D(a)$ remains a challenging task.

Fig. 3 (a) and (b) show the distributions of cascade sizes $D(s)$ for $\epsilon = 0.001$ and 0.005. The key features are: (i) $D(s)$ shows large fluctuations for neighboring values of s even after averaging over 500 realizations, in sharp contrast with $D(a)$; and (ii) the trend of $D(s)$ is similar to $D(a)$, with a rise at small s to a peak and then drops with a longer tail. These features can be explained by an approximation similar to the one developed for $D(a)$.

Let a_l and a_r again denote the micro time steps that a cascading block ceases to expand on both directions in a cascade. For simplicity, we ignore the dissipation of grains that fall back into the interior nodes in an expanding block and dissipation enters into the consideration only for grains falling onto the exterior nodes at the two ends. Under this assumption, the cascade size $s = a_l \cdot a_r$. The example in Fig. 1 is one in which $a_l = a_r = 3$ and there are 9 times (see red diamonds in the micro time steps) that some node has toppled before the system retains its stability. Note that a node may topple multiple times. We further remark that the relation $s = a_l \cdot a_r$ is generally valid. We illustrate an example in Fig. 1 had a_l been $a_l = 4$ instead of $a_l = 3$. Three more cascades would occur at the nodes marked with a triangle inside the closed blue circles just beyond the left boundary of the affected region corresponding to $a_l = 3$ (see micro time steps 4, 5 and time step $t + 1$). The cascade size is again given by $a_l \cdot a_r = 12$ and the cascade area by $a_l + a_r - 1 = 6$. These relations can be traced back to the geometry of the region affected when a grain is dropped, reviewed as a function of time. For the example of $a_l = 3$ and $a_r = 3$ (see red diamonds), the affected region forms a parallelogram from micro time steps 1 to 5. The number of distinguishable nodes being affected is $a_l + a_r - 1$ and the number of cascades is given by the product of the two sides of the parallelogram, namely $a_l \cdot a_r$. With this assumption and ignoring spatial correlations as before, an approximation to the distribution of cascade sizes is

$$\widetilde{D}(s) = \phi_0 \delta_{s,0} + (1-\phi_0)\epsilon_{eff}^2 \sum_{a_l \cdot a_r = s} (1-\epsilon_{eff})^{a_l + a_r - 2}, \tag{5}$$

where the summation is over all the cases where the product of two numbers a_l and a_r gives the required value of s.

Using Eq. (1) for ϵ_{eff}, Eq. (5) gives $\widetilde{D}(s)$ in terms of ϕ_0 and ϵ but it does not allow a closed form. For a given s, we can evaluate $\widetilde{D}(s)$ numerically by using ϕ_0 extracted from simulations and the results are shown in Fig. 3(c) and (d) on the same scales as the actual $D(s)$. The results show that $\widetilde{D}(s)$ in Eq. (5) is a reasonable approximation to $D(s)$ and gives the correct trend. More importantly, $\widetilde{D}(s)$ captures the fluctuations as observed in $D(s)$ and serves to illustrate that these fluctuations are intrinsic and they cannot be averaged away. Inspecting Eq. (5) gives two factors for these fluctuations. Firstly, the summands in Eq. (5) spread over a range that depends sensitively on the value of s. For a given s, writing s as the product of 1 and s gives $a_l + a_r - 2 = s - 1$ and a smallest summand $(1-\epsilon_{eff})^{s-1}$. On the other hand, if s can be factorized into a product of nearly identical a_r and a_l, then $a_r + a_l$ is smallest making $(1-\epsilon_{eff})^{a_r+a_l-2}$ the biggest. Thus, the summands in general range between

$$(1-\epsilon_{eff})^{s-1} \le (1-\epsilon_{eff})^{a_l+a_r-2} \le (1-\epsilon_{eff})^{2\sqrt{s}-2}. \tag{6}$$

Secondly, the number of ways that a certain value of s factorizes into a product of two integers can vary by much for two nearby values of s_1 and s_2. This implies that the number of terms in the summation in Eq. (5) differ drastically even between two nearby values of s. For instance, the summand takes on only the lower bound if s is a prime number. Putting the two factors together, both the number of terms in the summation as well as the values of the terms to be summed up can differ by much for two

UNCORRECTED PROOF

Please cite this article in press as: J. Liu, P.M. Hui, Areas and sizes of cascades in dissipative one-dimensional sandpile model, Phys. Lett. A (2017), http://dx.doi.org/10.1016/j.physleta.2017.05.022

nearby s_1 and s_2. Thus, the resulting $\widetilde{D}(s_1)$ and $\widetilde{D}(s_2)$ can change dramatically, resulting in the observed fluctuations in $D(s)$.

Despite capturing successfully the origin of the fluctuations in $D(s)$, the approximated $\widetilde{D}(s)$ retains the fluctuations to large s while the fluctuations in the actual $D(s)$ are suppressed for large s. The reason is that we have ignored the dissipation of the grains that fall into the interior of the growing block so that $s = a_l \cdot a_r$ works and Eq. (3) can be written down. For small s, the number of micro time steps for the system to settle is small and the assumption works better. For large s, there are many times that grains fall back inside the block and the assumption becomes worse.

4. Discussion

We close with a discussion on ϕ_0, which is the only parameter needed in $\widetilde{D}(a)$ and $\widetilde{D}(s)$. So far, we extracted ϕ_0 from simulation results and used it in ϵ_{eff} and approximating $D(a)$ and $D(s)$. Here, we develop a semi-empirical expression for ϕ_0 that is in excellent agreement with simulation results. Although the number of grains in the system varies from time step to time step, on average the number of grain dropped on to the system equals the number of grains dissipated [10]. The average cascade size S is thus

$$S = \frac{1}{2\epsilon}. \tag{7}$$

On the other hand, the cascade area is given by $a = a_l + a_r - 1$. Including dissipation only for grains falling to exterior nodes in an expanding block gives the size $s = a_l \cdot a_r$. Thus, for a given cascade area a, the possible cascade sizes take on discrete values with an average

$$\bar{S}_a = \frac{1}{6}(a+1)(a+2)\,, \tag{8}$$

where the subscript a emphasizes the condition of a given cascade area a. Averaging both sides over the distribution $\widetilde{D}(a)$ gives

$$\langle \widetilde{S} \rangle = \frac{1}{6}\langle a^2 \rangle + \frac{1}{2}\langle a \rangle + \frac{1}{3} \tag{9}$$

where analytic expressions for $\langle a \rangle$ and $\langle a^2 \rangle$ can be derived using Eq. (4) and they are functions of ϵ_{eff} and thus ϕ_0. Note that $\langle \widetilde{S} \rangle > S$ because dissipation of the grains that fall on to the interior nodes of a block would lower the actual cascade sizes. Hence, we assume that $\langle \widetilde{S} \rangle = K \cdot S$ with $K > 1$. It follows from Eq. (9) that

$$K \cdot S = \frac{\epsilon_{eff}^2}{(1-\epsilon_{eff})(-\ln(1-\epsilon_{eff})^3)}\left(1 - \frac{1}{\ln(1-\epsilon_{eff})}\right) + \frac{1}{3}\,, \tag{10}$$

which connects S and ϵ_{eff}. Finally, combining Eq. (7) and Eq. (10) gives an equation of K in terms of ϵ_{eff} and ϵ as

$$K = \frac{2\epsilon\epsilon_{eff}^2}{(1-\epsilon_{eff})(-\ln(1-\epsilon_{eff})^3)}\left(1 - \frac{1}{\ln(1-\epsilon_{eff})}\right) + \frac{2\epsilon}{3}\,. \tag{11}$$

Once K is known, Eq. (11) can be solved for ϵ_{eff} and thus ϕ_0. After examining simulation results of ϕ_0 for many different dissipation parameters ϵ, we found that one single value of $K = 4$ gives ϕ_0 that are in excellent agreement with simulation results over a wide range of ϵ from 0.001 to 0.019. The results are shown in Fig. 4. Eq. (11) thus provides a semi-empirical formula that gives ϵ_{eff} and hence ϕ_0 for use in $\widetilde{D}(a)$ and $\widetilde{D}(s)$.

In summary, we studied the BTW sandpile model on a one-dimensional periodic chain. In the absence of boundary nodes, a dissipation mechanism is needed. The physics turns out to be rich. The results in the cascade area distribution $D(a)$ and cascade size distribution $D(s)$ show non-trivial features that ask for an explanation. The cascade area distribution $D(a)$ shows a rise

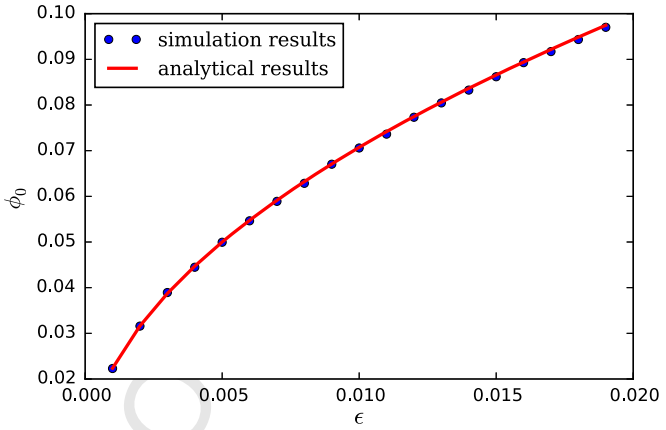


Fig. 4. Fraction of load-0 nodes ϕ_0 as a function of the dissipation probability ϵ. The semi-empirical formula (Eq. (11) with $K = 4$) gives results (line) that are in excellent agreement with simulation results (dots).

to a peak and a drop with a longer tail and $D(s)$ exhibits large fluctuations for nearby values of s. Exploiting the advantage that cascading processes can be analyzed in 1D systems, we developed approximations to the two distributions within an independent-site assumption. Ignoring spatial correlation of the states of neighboring nodes, an approximated $\widetilde{D}(a)$ was developed in terms of a single parameter ϕ_0. It captures the key features and gives the most probable value of cascade area correctly. The limitation of the theory also indicates that a better theory should include spatial correlation. By ignoring spatial correlation and dissipation of grains falling onto interior nodes of an expanding block, an approximated $\widetilde{D}(s)$ was developed again in terms of ϕ_0. The result illustrates that the fluctuations in $D(s)$ are intrinsic and they cannot be averaged away with different realizations. The approximated $\widetilde{D}(s)$ also gives a physically transparent explanation of the fluctuations, emphasizing the fact that the number of ways of getting at a given cascade size s varies widely even for two nearby values of s. A semi-empirical formula was found for the parameter ϕ_0 as a function of the dissipation parameter ϵ that agrees with numerical results very well. Being the only parameter in the approximations for $\widetilde{D}(a)$ and $\widetilde{D}(s)$, ϕ_0 can then either be extracted from simulation results or evaluated by the semi-empirical formula. Our work will be useful in understanding SOC in complex networks in that these networks typically have no boundary nodes for grains to fall out of the system in sandpile-type cascading processes. Therefore, a dissipation mechanism is often invoked in studying cascading processes in complex networks. The results here, therefore, sheds light on the non-trivialness of these cascading problems with dissipations by illustrating that even a one-dimensional chain exhibits results that can only be dealt with approximately.

References

[1] P. Bak, How Nature Works: The Science of Self-Organised Criticality, Copernicus Press, New York, NY, 1996.
[2] P. Bak, C. Tang, J. Geophys. Res. 94 (1989) 635.
[3] T. Chang, Phys. Plasmas 6 (1999) 4137.
[4] P. Bak, M. Paczuski, Physica A 348 (2005) 277.
[5] M. Bartolozzi, S. Drożdż, D. Leinweber, J. Speth, A. Thomas, Int. J. Mod. Phys. C 16 (2005) 1347.
[6] M. Paczuski, S. Boettcher, M. Baiesi, Phys. Rev. Lett. 95 (2005) 181102.
[7] B.A. Carreras, D.E. Newman, I. Dobson, A.B. Poole, IEEE Trans. Circuits Syst. I, Regul. Pap. 51 (2004) 1733.
[8] A. Deluca, N.R. Moloney, A. Corral, Phys. Rev. E 91 (2015) 052808.
[9] H.J. Jensen, Self-Organized Criticality: Emergent Complex Behavior in Physical and Biological Systems, Cambridge University Press, 1998.
[10] K. Christensen, N.R. Moloney, Complexity and Criticality, Imperial College Press, 2005.
[11] P. Bak, C. Tang, K. Wiesenfeld, Phys. Rev. Lett. 59 (1987) 381.
[12] P. Bak, C. Tang, K. Wiesenfeld, Phys. Rev. A 38 (1988) 364.

UNCORRECTED PROOF

Please cite this article in press as: J. Liu, P.M. Hui, Areas and sizes of cascades in dissipative one-dimensional sandpile model, Phys. Lett. A (2017), http://dx.doi.org/10.1016/j.physleta.2017.05.022

[13] D. Dhar, R. Ramaswamy, Phys. Rev. Lett. 63 (1989) 1659.
[14] D. Dhar, Phys. Rev. Lett. 64 (1990) 1613–1616.
[15] Y.-C. Zhang, Phys. Rev. Lett. 63 (1989) 470–473.
[16] K. Christensen, Z. Olami, Phys. Rev. E 48 (1993) 3361.
[17] D. Dhar, Physica A 263 (1999) 4.
[18] B. Drossel, Phys. Rev. E 61 (2000) R2168(R).
[19] R. Dickman, J.M.M. Campelo, Phys. Rev. E 67 (2003) 066111.
[20] P.-A. Noël, C.D. Brummitt, R.M. D'Souza, Phys. Rev. Lett. 111 (2013) 078701.
[21] A.-L. Barabási, R. Albert, Science 286 (1999) 509.
[22] L.A.N. Amaral, A. Scala, M. Barthelemy, H.E. Stanley, Proc. Natl. Acad. Sci. USA 97 (21) (2000) 11149–11152.
[23] D.J. Watts, S.H. Strogatz, Nature 393 (1998) 440.
[24] J. Gómez-Gardeñes, Y. Moreno, Phys. Rev. E 73 (2006) 056124.
[25] E. Bonabeau, J. Phys. Soc. Jpn. 64 (1995) 327.
[26] S. Lise, M. Paczuski, Phys. Rev. Lett. 88 (2002) 228301.
[27] L. de Arcangelis, H. Herrmann, Physica A 308 (2002) 545.
[28] A. Scala, V. Zlatić, G. Caldarelli, G. D'Agostino, Eur. Phys. J. Spec. Top. 225 (2016) 2017–2023.
[29] K.I. Goh, D.S. Lee, B. Kahng, D. Kim, Phys. Rev. Lett. 91 (2003) 148701.
[30] D.-S. Lee, K.-I. Goh, B. Kahng, D. Kim, Physica A 338 (2004) 84.
[31] C.D. Brummitt, R.M. D'Souza, E.A. Leicht, Proc. Natl. Acad. Sci. USA 109 (2012) E680.
[32] O. Malcai, Y. Shilo, O. Biham, Phys. Rev. E 73 (2006) 056125.
[33] C.-Y. Lin, C.-F. Chen, C.-N. Chen, C.-S. Yang, I.-M. Jiang, Phys. Rev. E 74 (2006) 031304.
[34] H. Bhaumik, S.B. Santra, Phys. Rev. E 88 (2013) 062817.
[35] H. Bhaumik, S.B. Santra, Phys. Rev. E 94 (2016) 062138.
[36] J. Lahtinen, J. Kertész, K. Kaski, Physica A 349 (2005) 535.

UNCORRECTED PROOF

Please cite this article in press as: J. Liu, P.M. Hui, Areas and sizes of cascades in dissipative one-dimensional sandpile model, Phys. Lett. A (2017), http://dx.doi.org/10.1016/j.physleta.2017.05.022